INCH-POUND
MIL-T-27/36C
8 June 2012
SUPERSEDING
MIL-T-27/36B
13 March 1991

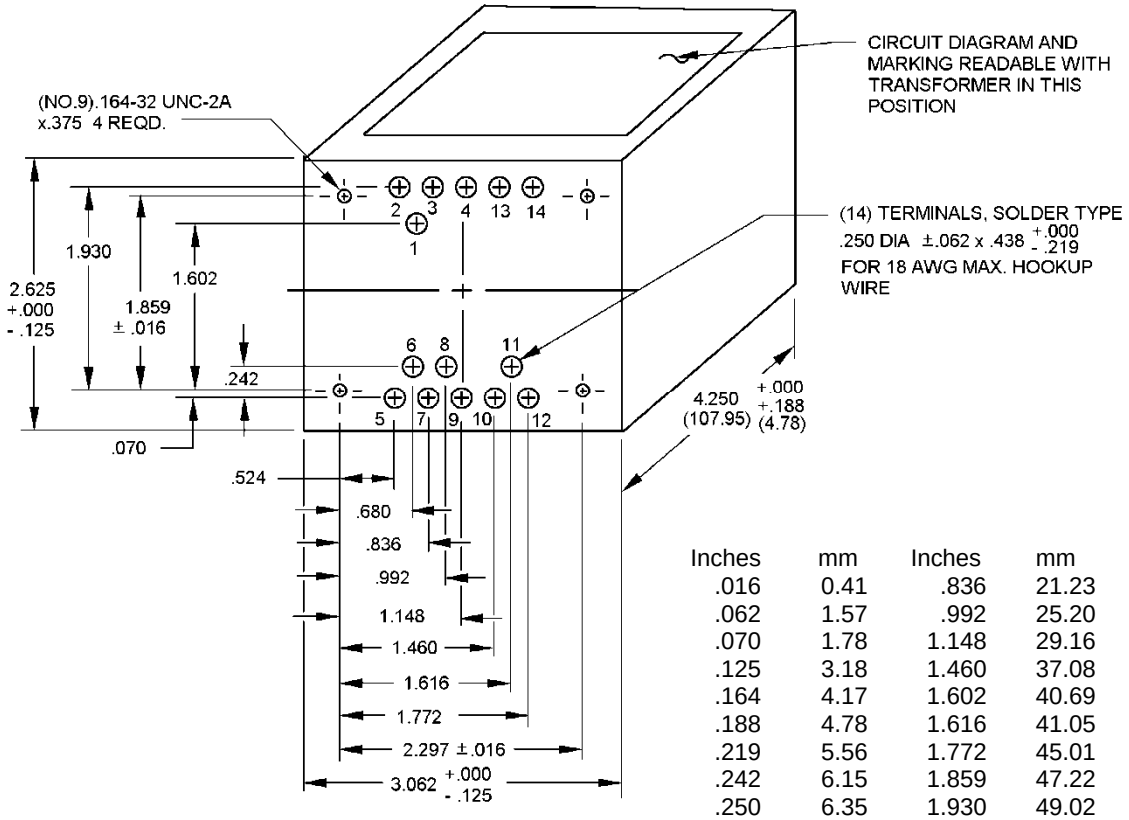
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP DOWN AND STEP UP

Inactive for new design after 13 March 1991.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.



Inches	mm	Inches	mm
.016	0.41	.836	21.23
.062	1.57	.992	25.20
.070	1.78	1.148	29.16
.125	3.18	1.460	37.08
.164	4.17	1.602	40.69
.188	4.78	1.616	41.05
.219	5.56	1.772	45.01
.242	6.15	1.859	47.22
.250	6.35	1.930	49.02
.375	9.53	2.297	58.34
.438	11.13	2.625	66.68
.524	13.31	3.062	77.77
.680	17.27	4.250	107.95

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.031 (0.79 mm).

FIGURE 1. Dimensions and configurations.

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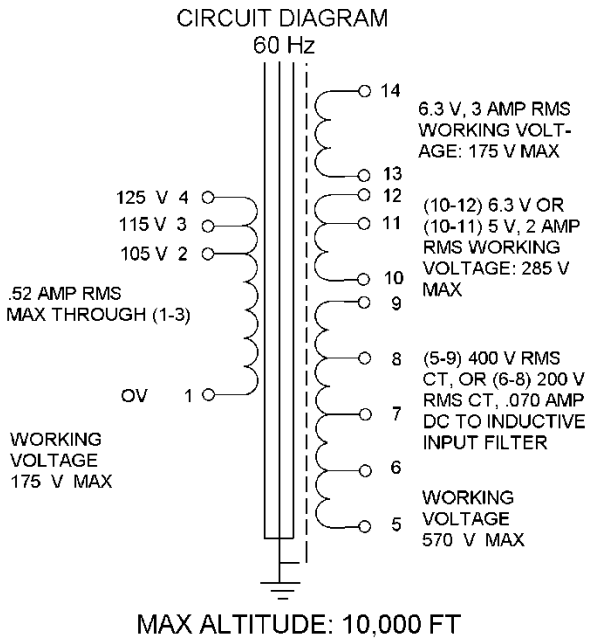

FIGURE 2. Circuit diagram.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-4) are used, they indicate the extreme terminals of the windings.)

Electrical ratings:

Primary voltage (1-2-3-4): 105/115/125 V, 60 Hz ±10 percent.

Primary current (1-3): 0.52 ampere rms maximum.

Secondary voltage:

(5-6-7-8-9): (5-9) 400 V rms, ct, or (6-8) 200 V rms, ct, 0.070 ampere dc to inductive input filter with full-wave rectifier.
(10-11-12): 6.3/5 V, 2 amperes.
(13-14): 6.3 V, 3 amperes.

Working voltage:

(1-4): 175 V maximum.
(5-9): 570 V maximum.
(10-12): 285 V maximum.
(13-14): 175 V maximum.

Design and construction.

Dimension and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal encased.

Material: Steel.

Altitude: 10,000 feet maximum.

Terminals: Solder lug, number 18 AWG.

Height: 0.438 +0.000, -0.219 inch.

Weight: 4.625 pounds maximum.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 4.5 pounds.

Dielectric withstanding voltage (at sea level):

(1-4): 1,500 V rms.
(5-9): 1,600 V rms.
(10-12): 1,600 V rms.
(13-14): 1,500 V rms.

Electrical characteristics:

No load: With 115 V and 60 Hz in (1-3).

Current in (1-3): 0.190 ampere rms maximum.

Power in (1-3): 2.6 watts maximum.

Voltage across (5-9): 437 V maximum.

Voltage across (6-8): 218 V maximum.

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Unbalance (5-7) and (7-9): 2 percent maximum.

Unbalance (6-7) and (7-8): 2 percent maximum.

Voltage across (10-11): 5.5 V maximum.

Voltage across (10-12): 6.89 V maximum.

Voltage across (13-14): 7.01 V maximum.

Voltage across (1-2): 105 V ±1 percent.

Voltage across (1-4): 125 V ±1 percent.

Rated load: With 105 V, 60 Hz across (1-2) and (5-9) loaded with 5R4WGB full wave rectifier and 0.070 ampere dc into 7.5 H inductive input filter and rated current in (10-12) and (13-14):

Voltage across (5-9): 400 V ±4 percent.

Voltage across (10-12): 6.3 V ±2 percent.

Voltage across (13-14): 6.3 V ±2 percent.

Electrostatic shielding: Voltage ratio: 5 to 1 at 20 kHz.

Polarity: Additive, with terminals 4 and 5; 9 and 10; and 12 and 13 connected.

Temperature rise: +40°C with 105 V, 54 Hz across (1-2) and (5-9) loaded with 5R4WGB full wave rectifier and 0.070 ampere dc into 7.5 H inductive input filter and rated current in (10-12) and (13-14) at an ambient temperature of +65°C.

VERIFICATION:

Qualification inspection: Not applicable.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/36-01.

NOTE: Transformers covered by this specification sheet are substitutable for transformers, MS part number MS90026-2.

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2009-085)

Review activities:
Army – AR, CR4, MI
Navy – OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.daps.dla.mil.

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